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Key Points:

- The dominant sources of neodymium to the Atlantic Ocean are rivers and dust, which means it is best characterized as a “top-down” system
- The dominant source of neodymium to the Pacific Ocean is from sediments at all depths which means it is best characterized as a “bottom-up” system
- Neodymium isotopes can be used as a proxy for water-mass mixing in the Atlantic Ocean and a proxy for residence time in the Pacific Ocean

Supporting Information:

Supporting Information may be found in the online version of this article.

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Differing Controls on the Cycling of Neodymium in the Atlantic and the Pacific

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Abstract Neodymium (Nd) isotopes ratios are regarded as a key proxy for past ocean circulation despite unsettled debate on marine Nd cycle controls, which opposes a “top-down” hypothesis (surface sources dominate) and a “bottom-up” hypothesis (sedimentary sources dominate), each with important implications for Nd isotope proxy interpretation. We suggest a new, basin-specific framework for interpreting Nd isotope records. Specifically, while Nd isotope records are reliable water-mass-mixing proxies in the Atlantic, they are likely better used as proxies of residence time in the Pacific. Here, we test these hypotheses using a computationally efficient, data-constrained model of the marine Nd cycle. Our results show Atlantic Nd isotopes are dominated by surface sources, while Pacific Nd isotope ratios are strongly influenced by sedimentary sources. We find that shallow, reactive sediments—especially those with extremely radiogenic or unradiogenic isotopic compositions—are crucial to reproduce observed ϵNd profiles in the Pacific.

Plain Language Summary Neodymium isotopes are chemical tracers that scientists use to study how oceans circulated in the past. There has been a debate about where the majority of Nd enters the ocean and what controls its distribution. Some think it mostly comes from surface sources such as rivers and dust (“top-down”), while others believe it mainly comes from the seafloor and ocean sediments (“bottom-up”). This study uses a computer model of the ocean to test both ideas and finds that in the Atlantic Ocean, surface sources are more important—supporting the top-down idea. But in the Pacific Ocean, the seafloor plays a more significant role, especially shallow sediments that release large quantities of Nd. This means while Atlantic sediment records tell us more about water mass mixing, Pacific records inform us about how slowly or quickly deep water moves. Our findings show that the interpretation of Nd in the ocean is complex and that source-specific interpretations need to be considered when applying it to study past oceans.

1. Introduction

The deep ocean is the largest reservoir of carbon on the surface of the Earth, with a capacity approximately 60 times greater than the atmosphere (Sigman & Boyle, 2000). Therefore, changes to deep ocean carbon storage, which are strongly influenced by ocean circulation (Adkins, 2013), have a large impact on atmospheric pCO_2 and climate. Fully understanding how changes in ocean circulation have influenced glacial-interglacial climate relies on accurate paleoceanographic reconstructions from geochemical proxy measurements. One such proxy is the isotope ratio of neodymium (Nd) in seawater, reported as ϵNd , the $^{143}\text{Nd}/^{144}\text{Nd}$ in parts per 10,000 relative to an estimate of bulk Earth composition (Goldstein & Hemming, 2003; Jacobsen & Wasserburg, 1980). Unlike other circulation proxies, Nd is not biologically cycled and thus remains less sensitive to changes in productivity that also vary on glacial-interglacial timescales.

Seawater Nd is originally sourced from weathering and erosion of continental rocks, with isotopic compositions that vary as a function of age and lithology (Goldstein & Hemming, 2003; Robinson et al., 2021). The concentration of dissolved Nd ([Nd]) in the ocean is similarly influenced by these inputs but is also modulated by scavenging onto particles and reversible exchange with sinking particulate matter (Arsouze et al., 2007; Rempfer et al., 2011; Siddall et al., 2008). While ϵNd reflects the provenance of Nd sources, [Nd] provides insights into the balance between inputs (rivers, dust, sediments) and removal processes, making their combined analysis critical for understanding marine Nd cycling. The oldest rocks, with the most negative ϵNd values (termed “unradiogenic”) are found in Canada and Greenland, and they imprint seawater in the North Atlantic with signatures of -13 to -14 (Lambelet et al., 2016; Robinson et al., 2021). The youngest rocks, with the least negative ϵNd values

(termed “radiogenic”) are found around the Pacific rim, and they imprint seawater in the North Pacific with signatures of -3 to -4 (Wu, 2019). To first approximation, ϵNd appears to mix conservatively in the ocean interior, which has led to its use as a water-mass mixing tracer.

The primary inputs of Nd to the ocean are via rivers, dust, and sediments and Nd is redistributed within the ocean's interior via circulation and reversible scavenging (Arsouze et al., 2007; Du et al., 2020; Pöppelmeier et al., 2020; Rempfer et al., 2011; Tachikawa et al., 2003). Initially, rivers and dust were thought to dominate Nd sources (Frank, 2002; Goldstein & Hemming, 2003). This top-down configuration assumes that interior sources and sinks are negligible and that ϵNd mixes conservatively, making it an effective water-mass-mixing tracer. Then, it was proposed that continental margin “boundary exchange” (Jeandel, 2016; Jeandel et al., 2007) was also a significant contributor of [Nd] to the global Nd budget, calling into question the conservativeness of ϵNd as a water mass tracer. More recently, new observational data have supported the idea that abyssal (Abbott et al., 2015; Blaser et al., 2020; Du et al., 2020; Robinson et al., 2023) sources are also important contributors to the marine Nd cycle, challenging the top-down hypothesis in favor of a bottom-up regime of Nd cycling. A bottom-up regime precludes interpreting ϵNd as a conservative water-mass mixing proxy since ϵNd changes as a Nd flux from sediments is continuously added along a water mass' flow path. However, in this regime, ϵNd could be used as a residence-time tracer, as a slower circulation allows more time for Nd isotopes to accumulate in a given water mass and modify its isotopic composition (Du et al., 2018, 2025; Hu & Piotrowski, 2018). Determining which regime applies when and where will dramatically improve the utility of ϵNd as a paleoceanographic tracer.

2. Methods

We used a steady-state model of the marine Nd cycle, the Global Neodymium Ocean Model (GNOM; Text S1 in Supporting Information S1; Pasquier, Hines, et al., 2022; Pasquier, Primeau, et al., 2022) to systematically test the top-down and bottom-up hypotheses. In GNOM, the [Nd] and ϵNd model is embedded in the data-constrained Ocean Circulation Inverse Model (OCIM2; DeVries & Holzer, 2019) which has $2^\circ \times 2^\circ$ horizontal and 24 layer resolution. This spatial resolution is similar to that of typical climate models (Arsouze et al., 2007; Gu et al., 2019; Rempfer et al., 2011; Tachikawa et al., 2003) but affords computational efficiencies rivaling simple box models. There are five sources of Nd to the ocean in GNOM: mineral dust, volcanic ash, sediments, rivers, and submarine groundwater, however volcanic ash and submarine groundwater constitute less than 0.1% of the total so we focus on dust, sediments, and rivers in this study. Neodymium is also internally cycled via reversible scavenging onto four particle types (Text S1 in Supporting Information S1) and carried by the ocean circulation. We first optimized GNOM to match global [Nd] and ϵNd observations (deemed “Best Fit”; Figure 1; Text S2 in Supporting Information S1). Then, we performed a variety of model experiments to determine which sources are most important for controlling the global [Nd] and ϵNd distributions and how each source impacts the [Nd] and ϵNd profiles in different ocean basins.

We improved on the work by Pasquier, Hines, et al. (2022) by enforcing more realistic parameter ranges. In GNOM, parameters can be optimized to minimize model–data mismatches (Text S2 in Supporting Information S1). Here we highlight key changes between GNOMv1 and GNOMv2, and provide more details in Text S3 in Supporting Information S1. As laboratory studies constrain Nd solubility in dust to $\sim 2\%$ (Greaves et al., 1994), our β_{dust} range has been reduced to 0%–5%. Relatedly, the Nd concentration in dust was decreased from 40 to 27 $\mu\text{g/g}$ (Goldstein et al., 1984). While GNOM can simulate hydrothermal Nd fluxes, observational evidence suggests that hydrothermal vents may act as a net sink for Nd (Basak et al., 2024), so hydrothermal sources were excluded. Finally, volcanic ash ϵNd values were restricted to be consistent with estimates of 0 to +10 (Kelemen et al., 2007).

To assess how mineral dust, rivers, and sedimentary input of Nd impacts the distribution of Nd isotopic composition, we reoptimized GNOM but with each source omitted (Text S4 and Figure S1 in Supporting Information S1), then used the optimized parameters from the Best Fit run and systematically withheld each of them in turn (“Source Subtraction”; Figures 2a–2l).

To determine whether surface or deep Nd sources are more important for setting ϵNd and [Nd] distributions, we pay particular attention to the sedimentary source—testing the effect of different sedimentary source profiles as a function of water depth. In GNOM, this profile is controlled by 3 parameters: the sedimentary flux at depth (ϕ_∞), the flux at the surface of the ocean (ϕ_0), and the depth attenuation (z_0), which controls the root of the exponential function that governs the sedimentary flux of Nd (see also Text S1). Note that, while studies such as Arsouze

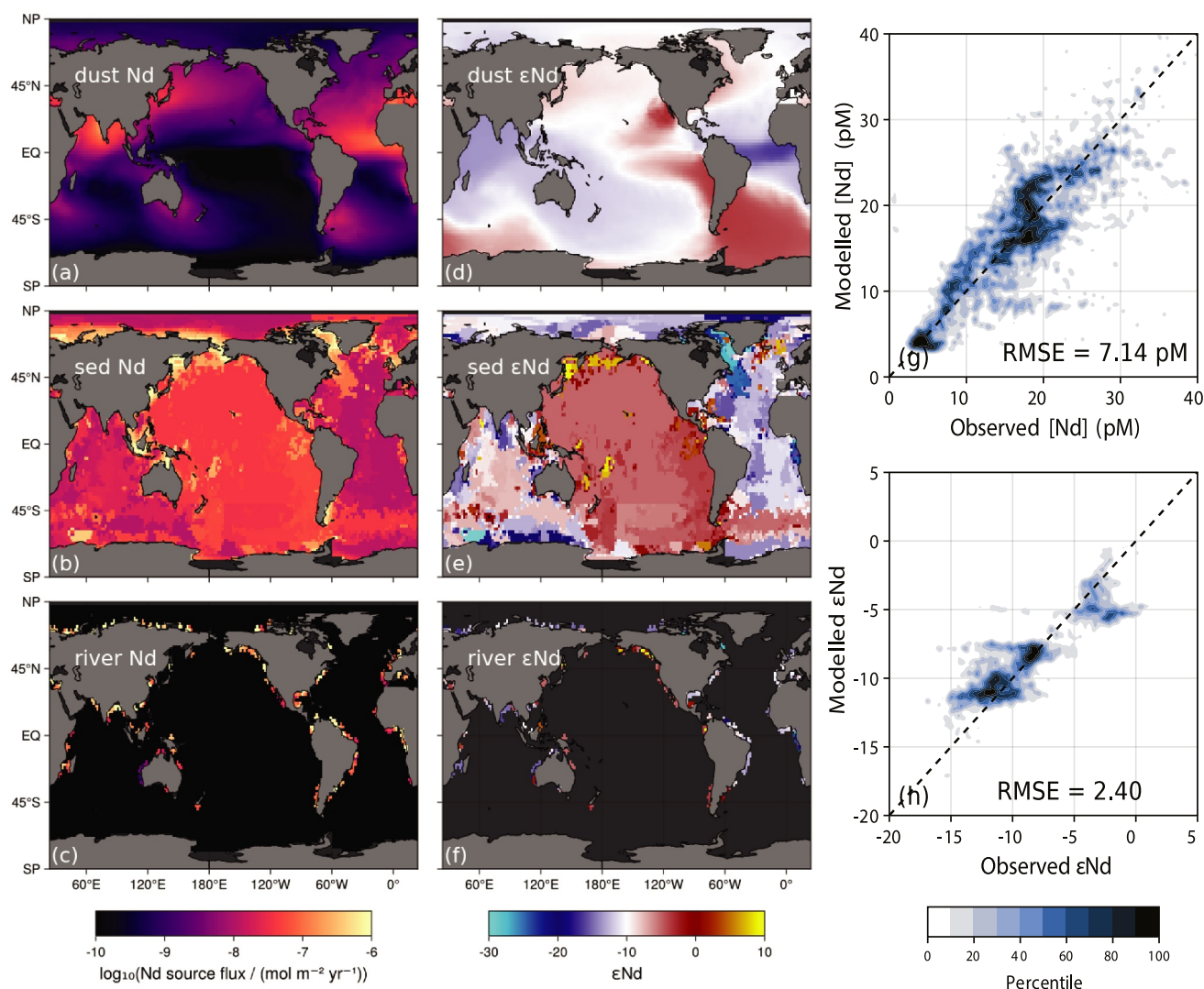


Figure 1. The GNOMv2 Best Fit model. The magnitude of the Nd source from dust (a), sediments (b), and rivers (c) in $\text{mol m}^{-2} \text{yr}^{-1}$. The isotopic composition of Nd from dust (d), sediments (e), and rivers (f) in parts per ten thousand. In the Best Fit model, the most significant source of Nd to the oceans is sediments followed by rivers then dust. Joint density of observed and modeled [Nd] (g) and ϵNd (h), plotted such that $n\%$ of data lies outside the n th percentile contour.

et al. (2009) and Rempfer et al. (2011) distinguish between shallow continental margin (“boundary exchange”) and abyssal sedimentary sources, we model a single, depth-dependent source.

The sedimentary source is additionally scaled by “reactivity,” which is a parameterized parabolic function of sediment ϵNd that acts to enhance the release of Nd from sediments with extremely positive or negative ϵNd values. This enhanced release is motivated by the well-known observation that highly radiogenic young volcanic material is easily weathered (Blaser et al., 2016, 2020; Lacan & Jeandel, 2005; Pearce et al., 2013; Wilson et al., 2013). While this is not necessarily true for rocks with highly unradiogenic compositions, the Greenland Ice Sheet produces a large amount of freshly ground fine-grained glacial flour from rocks with extremely negative ϵNd compositions. The use of a reactivity parameter can be found in other ϵNd modeling studies (e.g., Poppelmeier et al., 2020).

To enforce a strictly bottom-up scenario for sedimentary Nd input, we ran an optimization where we required the sedimentary flux to linearly increase with depth (the “Deep Sediments Only”; Figure S3b in Supporting Information S1). This was accomplished by setting $z_0 = 50,000 \text{ m}$, $\phi_0 = 0 \text{ pmol cm}^{-2} \text{yr}^{-1}$, and allowing ϕ_∞ to vary. We also tested a “Shallow Sediments Only” scenario by fixing $z_0 = 300 \text{ m}$ and setting $\phi_\infty = 0 \text{ pmol cm}^{-2} \text{yr}^{-1}$. In

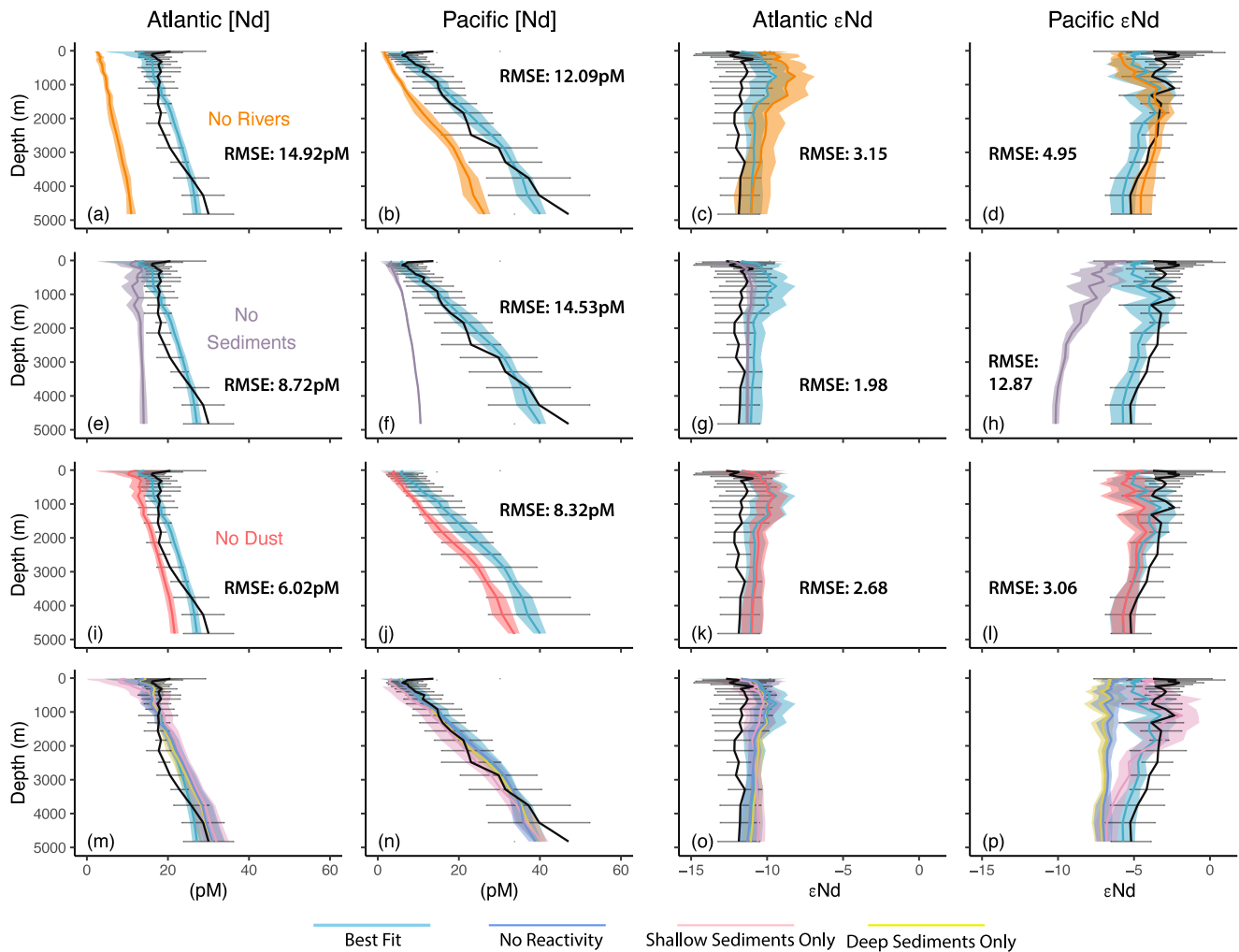


Figure 2. Comparison of the Best Fit model with Source Subtraction scenarios and sedimentary flux manipulation. The [Nd] RMSE of the Best Fit run (blue) is 6.06 and 8.34 pM for the Atlantic and Pacific, respectively. The RMSE of the Atlantic ϵ Nd and Pacific ϵ Nd is 2.35 and 2.84 (a) Mean Atlantic [Nd] depth profile from the No Rivers experiment (yellow) overlaid over the Best Fit model (blue) and observations (black). Ribbons and error bars indicate standard deviation. Root Mean Square Error (RMSE) is indicated. Panel (b) As panel (a) but for the Pacific. Panels (c, d) As panels (a, b) but for ϵ Nd. Panels (e–h) As panels (a–d) but for the No Sediments experiment (purple). Panels (i–l) As panels (a, b) but for the No Dust experiment. Panels (m–p) As panels (a–d) but for the three reoptimized runs: Constant Reactivity, Shallow Sediments Only, and Deep Sediments Only.

this scenario, only sediments above 2,000 m contribute [Nd] to the ocean (Figure S3c in Supporting Information S1). To test the importance of variable sediment reactivity, we set the parameter α_a to zero, which flattens the parabolic scaling function to 1. This turns off the mechanism by which the Best Fit model optimizes enhanced sediment reactivity, and thus higher Nd fluxes, for sediments with more extreme ϵ Nd values. This simulation is named the “Constant Reactivity” model. We designed these scenarios to be idealistic in order to isolate specific mechanisms, not to simulate realistic conditions.

GNOM's steady-state matrix schema uniquely allows us to precisely track Nd sources and transport pathways. We performed diagnostic runs (as detailed in Pasquier, Hines, et al. (2022), Pasquier, Primeau, et al. (2022)) where two endmember oceanic regions were defined, and [Nd] and ϵ Nd contributions from each region were tracked using labeling tracers. Specifically, we tagged [Nd] and ϵ Nd of water leaving the first region, accounted for reversible scavenging and burial along the flow path, and untagged it upon entry into the second region. We used this method to assess how conservative the transport of Nd is between the endmember regions and diagnose any addition or removal of Nd along the way. It is the first time that this technique has been applied to investigate trace-metal isotope mixing.

3. Results and Discussion

3.1. Impact of Dust, Rivers, and Sediments on the Marine Nd Cycle

The optimized GNOMv2 Best Fit run successfully reproduces the observed distributions of ϵNd and $[\text{Nd}]$ (Figure 1), with global root mean square error (RMSE) values of 2.40 and 7.14 pM, respectively. Model performance is better in the Atlantic (ϵNd RMSE = 2.35) than in the Pacific (ϵNd RMSE = 2.84). The best-fit solution is characterized by a top-down sedimentary source profile where the flux per unit area is highest at the surface and decreases with a depth attenuation scale of 300 m (Figure S3a in Supporting Information S1). This results in 67% of global Nd being sourced from surface inputs (rivers and dust) combined with shallow sediments above 2,000 m (Table S2 in Supporting Information S1). Globally integrated, however, deep sedimentary fluxes (>2,000 m) constitute 52% of the total sediment-derived Nd due to the large area of the deep seafloor (Figure S5b in Supporting Information S1).

Neodymium isotopes can be scavenged onto four particles types in GNOM: biogenic silica, particulate organic matter, mineral dust, and a fourth fictitious tracer that is termed “precipitation” to account for particles that have sparse observational data but may be key to Nd cycling such as hydrothermal particles (Schlitzer et al., 2018) and iron-manganese oxides (Basak et al., 2024; Du et al., 2025; Lagarde et al., 2020; Schijf et al., 2015). In our Best Fit model, only 24.6% of scavenged Nd is scavenged onto biogenic silica, POC, and dust (Figure S12 in Supporting Information S1). This supports recent studies that identify authigenic phases, particularly manganese oxides, as the most important particle type for reversibly scavenging Nd in the water column (Basak et al., 2024; Du et al., 2025; Lagarde et al., 2020).

The absence of mineral dust input affects total $[\text{Nd}]$ but has a weak effect on ϵNd (Figures 2k and 2l). Rivers influence $[\text{Nd}]$ in the Atlantic and Pacific basins but exert a much stronger control on the ϵNd of the Atlantic, where their removal leads to a more radiogenic profile (Figures 2a–2d). This is likely a function of basin volume, as riverine inputs represent a proportionally larger flux to the smaller Atlantic. Both basins depend on sedimentary Nd supply, but the Pacific ϵNd profile is far more sensitive to its removal, becoming excessively unradiogenic without sediments, underscoring the greater importance of seafloor sources in the larger Pacific basin (Figures 2e–2h).

3.2. Sedimentary Source Profiles

To further evaluate the top-down and bottom-up hypotheses, we ran GNOM with four distinct optimized sedimentary release profiles (Figure S3a–S3d in Supporting Information S1). The Best Fit model follows a top-down profile, where the sedimentary Nd flux is highest at the surface and decreases exponentially with a depth attenuation scale (z_0) of 300 m (Figure S3a in Supporting Information S1). We compared this against an optimized “Deep Sediments Only” profile configured so that Nd fluxes increase linearly with depth (Figure S3b in Supporting Information S1), an optimized “Constant Reactivity” profile which removes the enhancement of sedimentary flux as a function of the sediments' ϵNd value (Figure S3d in Supporting Information S1), and an optimized “Shallow Sediments Only” profile which excludes all sediment-derived Nd input below 2,000 m (Figure S3c in Supporting Information S1).

We find that altering the sedimentary source profile has a negligible effect on Atlantic ϵNd values but has a large effect on Pacific ϵNd values. In the Deep Sediments Only run, the optimization increased the reactivity parameter (α_d) to compensate for the minimized release of Nd from shallow sediments, which disproportionately host the most extreme ϵNd values (Figure S5a in Supporting Information S1). However, this adjustment was insufficient, and the modeled Pacific ϵNd remained too unradiogenic. This result is corroborated by the close correspondence between the Deep Sediments Only and Constant Reactivity runs (Figure 2p), the latter of which also produces a Pacific that is too unradiogenic due to reduced input from radiogenic shallow sediments.

Thus, we find that the inclusion of the sediment reactivity parameter is critical for matching the observed isotopic composition of seawater in the Pacific (Figure 2p). While shallow and reactive sediments are essential for setting the Pacific ϵNd profile, deep sediments are also important. In the Shallow Sediments Only run, the Pacific ϵNd profile matches the Best Fit run in the upper 2,000 m but diverges below, becoming too unradiogenic in the deep ocean. Therefore, matching observations requires sedimentary Nd input in the Pacific at all depths and the enhanced reactivity parameter to account for the disproportionate influence of shallow, radiogenic sediments,

which comprise a small fraction of the seafloor area (Figure S5b in Supporting Information S1) but exert a major control on dissolved ϵNd .

3.3. Differing Controls in the Atlantic and Pacific Basin

The contrasting neodymium (Nd) cycling regimes of the Atlantic and Pacific oceans arise from differences in basin geometry, circulation, and sediment characteristics. The Pacific Ocean's relatively sluggish abyssal circulation, and more radiogenic sediments result in prolonged contact between deep waters and reactive sediments, promoting both Nd accumulation and modification of bottom-water ϵNd . Our calculated Nd residence times support this interpretation, with substantially longer residence times and higher area-normalized sedimentary fluxes in the Pacific compared to the Atlantic (Table S2 and S3 in Supporting Information S1). Our model simulations indicate that the Pacific is characterized by a bottom-up regime, where its ϵNd distribution requires significant sedimentary input at all depths; excluding deep sedimentary sources produces a mismatch with observed values, particularly in the deep ocean (Figure 2p). Conversely, the Atlantic operates as a top-down system, dominated by surface inputs (Table S3 in Supporting Information S1).

Diagnostic tracking of Nd sources confirms this dichotomy. Surface sources (rivers and dust) constitute 71% of Atlantic Nd fluxes. In contrast, most Nd enters the Pacific in the deep ocean, with 61% sourced from sediments below 2,000 m and 25% from sediments above 2,000 m (Table S3 in Supporting Information S1). The transport and mixing of these end-member sources (surface Atlantic and seafloor Pacific) can account for the majority of oceanic Nd and its isotopic composition (Figures 3c and 3d); contributions from outside of these endmember regions only cause variations of approximately ± 1 epsilon unit (Figures 3e and 3f).

Our characterization of the distinct roles for sedimentary and surface fluxes of Nd to the Atlantic and Pacific Basins differ from previous descriptions of the marine Nd cycle, which suggest that large abyssal sedimentary sources in the global ocean control Nd isotope distributions (Abbott et al., 2015; Du et al., 2020, 2025; Haley et al., 2017). Early modeling work by Jones et al. (2008) successfully simulated interior Atlantic ϵNd through conservative mixing of surface ϵNd distributions, but they could not reconcile Pacific observations without invoking additional non-conservative processes, hinting at a fundamental inter-basin difference in cycling, though this study did not simulate Nd concentrations. Our inferred interpretation of bottom-up Nd cycling in the Pacific provides a mechanistic explanation for the non-conservative behavior of Pacific ϵNd (Jones et al., 2008) and is strongly supported by the recent analysis of Du et al. (2025), who identified benthic fluxes as the dominant control in the Pacific, but did not comment on Atlantic ϵNd controls. Our analysis supports a new conceptual framework for understanding the Nd isotope system that is basin-specific. The Nd cycle principally follows a bottom-up regime in the Pacific but a top-down regime in the Atlantic.

3.4. Application of Authigenic ϵNd as a Paleooceanographic Tracer

Our experiments with GNOM improve our understanding of the processes that control global Nd cycling and can improve our interpretations of ϵNd as a paleooceanographic tracer. To test where ϵNd can be utilized as a water-mass mixing proxy or a residence time proxy, we ran an idealized experiment where ocean transport was slowed down by 25% everywhere, while Nd sources were left untouched. While such a change is not realistic, it is nevertheless useful to explore as this experiment retains the same water mass mixing proportions while increasing residence times. We found that slowing the circulation changed ϵNd less in the Atlantic than in the Pacific (Figure S9 in Supporting Information S1), supporting the use of ϵNd as a water-mass tracer in the Atlantic and a proxy for residence time in the Pacific.

We calculated the North Atlantic Deep Water (NADW) fraction in each Atlantic grid cell by identifying an NADW end-member value ($\epsilon\text{Nd}_{\text{NADW}} = -13.5$, $[\text{Nd}_{\text{NADW}}] = 17.5$ pmol/kg) and an Antarctic Bottom Water (AABW) end-member value ($\epsilon\text{Nd}_{\text{AABW}} = -9.5$, $[\text{Nd}_{\text{AABW}}] = 20.0$ pmol/kg; Howe et al., 2016), for modern and slowed circulations (Figures 4a–4c). As expected, in the top-down Atlantic, the %NADW distribution is robust to the changed circulation. The earliest applications of the authigenic ϵNd tracer was to evaluate changes in North Atlantic water-mass structure during glacial/interglacial transitions (Pöppelmeier et al., 2018, 2021; Roberts et al., 2010; Roberts & Piotrowski, 2015; Zhao et al., 2019), and our top-down characterization of the Atlantic Nd regime supports this approach.

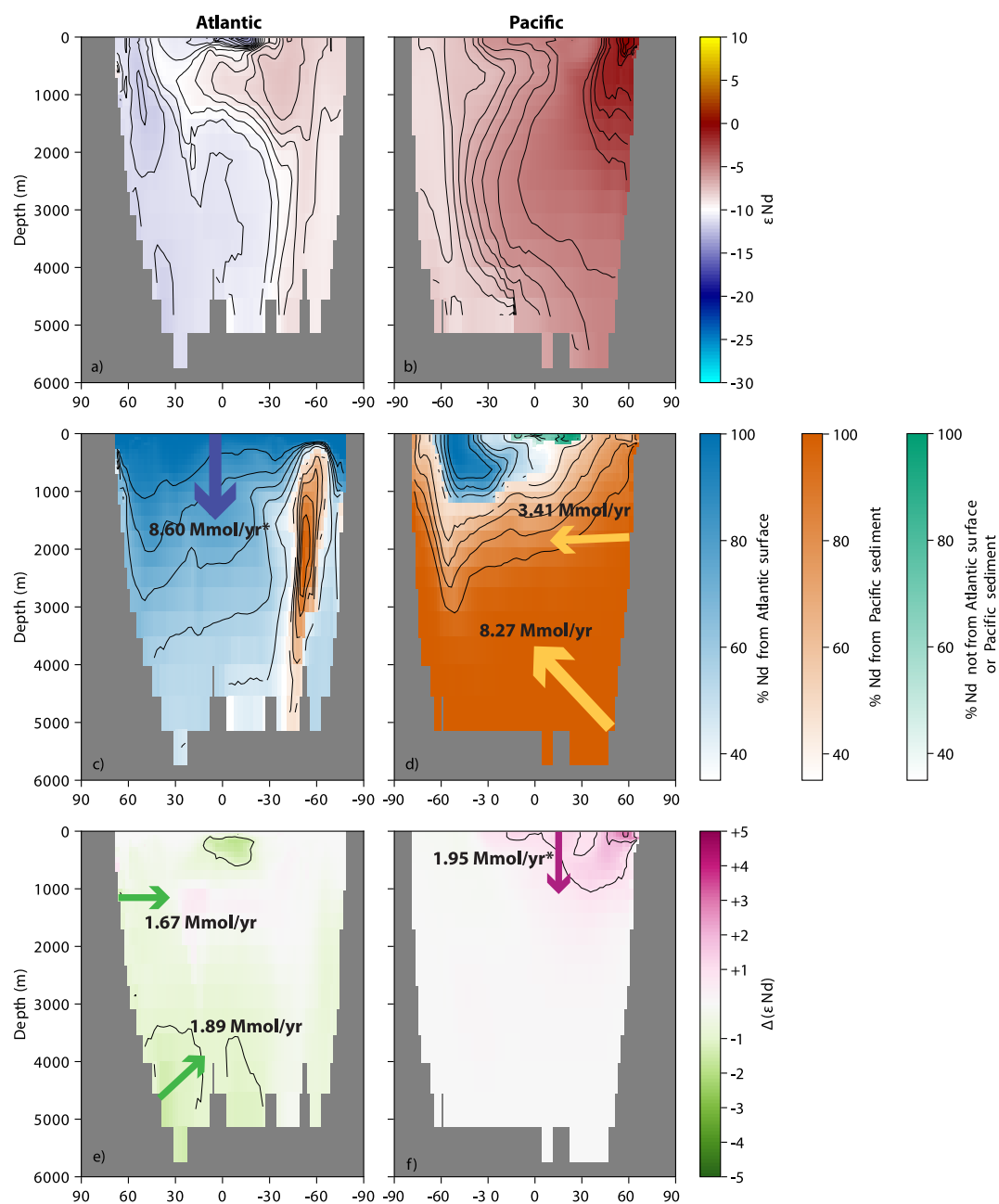


Figure 3. Nd source partitioning. (a) Zonal mean Atlantic ϵNd . Panel (b) As panel (a) but for the Pacific. Panels (c, d) As panels (a, b) but for the dominant fraction of Nd coming from either the Atlantic surface (blue), Pacific sediments (orange), or other (green). Panels (e, f) As panels (a, b) but for the change in ϵNd between the Best Fit model and a run with only Atlantic surface and Pacific sediment sources. Annotated arrows indicate Nd fluxes from the surface and from sediments.

We evaluated the potential of ϵNd as a proxy for water-mass age by computing the ideal age, which is the mean time since a water parcel was last in contact with the surface (England, 1995) and is easily computed using our OCIMv2 transport matrix (Text S7 in Supporting Information S1). Slowing the circulation shows that Atlantic ϵNd values are largely independent of residence time (Figure S9 in Supporting Information S1), whereas Pacific ϵNd responds strongly, displaying a clear positive linear relationship between water-mass age and ϵNd below 2,000 m for the modern circulation (Figure 4e).

In the deep Pacific, ϵNd varies linearly with ideal age, and this relationship is robust to changes in the circulation (Figures 4f and 4d). However, this relationship breaks down in the upper ocean, likely due to input from highly

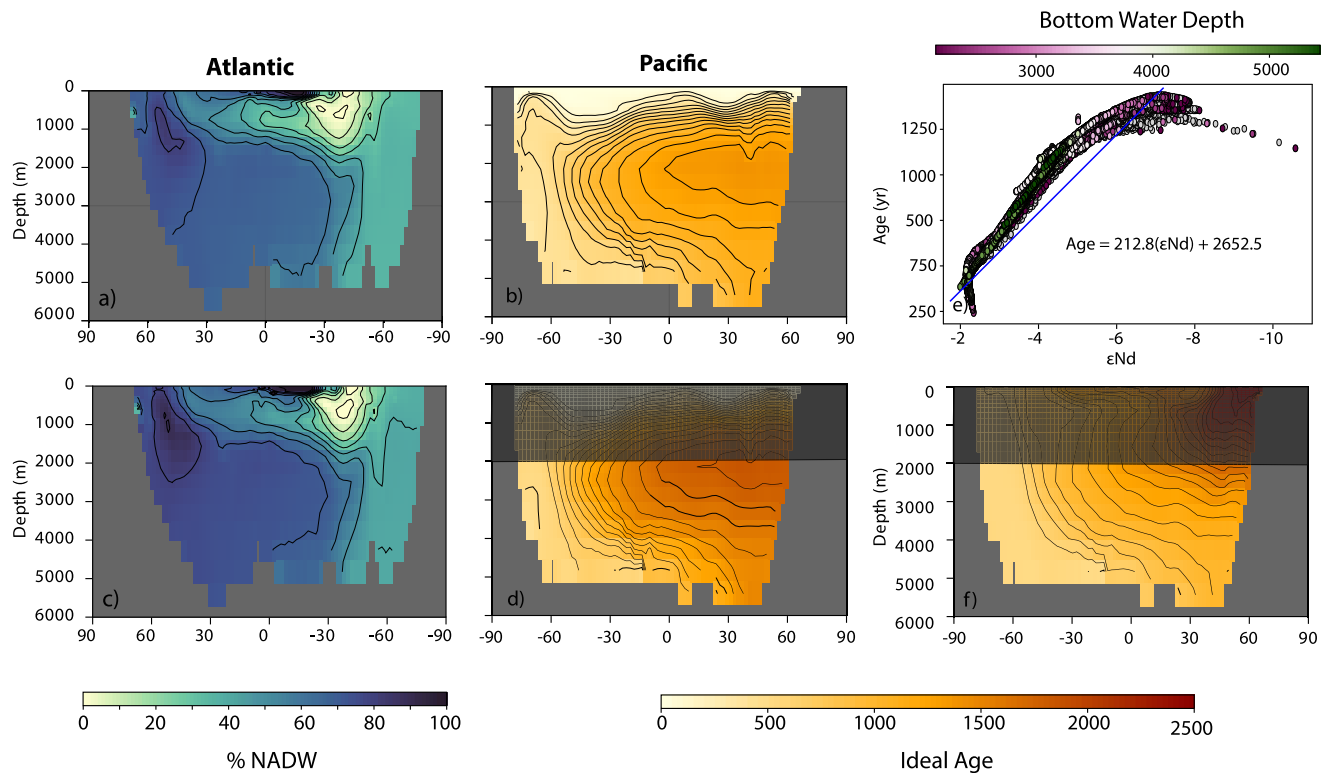


Figure 4. ϵ Nd proxy use for Atlantic and Pacific basin. (a) Fraction of NADW (% NADW) in each grid cell of the Atlantic Ocean. Panel (c) As panel (a) but for the circulation slowed down by 25%. Panels (b, d) As panels (a, c) but for the age of water in the Pacific basin. (e) Pacific Ocean ϵ Nd values plotted against ideal age for waters deeper than 2,000 m with the corresponding linear regression indicated. Ideal age is the mean time since a water parcel was last in contact with the surface (England, 1995) panel (f) As panel (d) but for the ideal age predicted by the linear regression shown in panel (e) for the slowed circulation. The upper 2,000 m of panels d and f are shadowed because the linear regression fails to capture the ϵ Nd/ideal age relationship above 2,000 m.

reactive, radiogenic sources on the shelf and continental slope (Figure 4e, Figure S5a in Supporting Information S1). This supports the application of ϵ Nd as a proxy for deep-ocean residence time in the Pacific (Du et al., 2018; Hu et al., 2016), and it is consistent with bottom-up control of ϵ Nd in the Pacific. We emphasize that this experiment is idealized; a full-fledged glacial simulation would also require changes in riverine inputs, sea level variation, dust flux, volcanic inputs, and scavenging regimes as well as potential shifts in ϵ Nd endmember values due to migration of convection zones (Meniel et al., 2020). More realistic simulations, including embedding transport matrices derived from past ocean circulations into GNOM, have the potential to refine the ϵ Nd–age relationship and improve Pacific ventilation reconstructions from ϵ Nd records.

Limited ϵ Nd data from the Indian and Southern Oceans preclude a definitive classification of their Nd-cycling regimes, though the relationship between ideal age and ϵ Nd was explored in this study (Figure S10 in Supporting Information S1). Approximately half of the Nd input to the Indian Ocean is from deep sediments, but there are also substantial fluxes from rivers and dust (Figure 1; Table S3 in Supporting Information S1). The Southern Ocean has isotopically diverse sediments, which could drive a large deep sedimentary flux (Figure 1; Table S3 in Supporting Information S1), and it has the smallest river and dust fluxes of any ocean basin, however the rapid circulation of the Antarctic Circumpolar Current may limit the sedimentary control of ϵ Nd distributions in this region. Additional [Nd] and ϵ Nd observations and further analysis are therefore required to determine whether the Indian and Southern Oceans are dominated by conservative water-mass mixing.

Our findings should be interpreted in light of GNOM's resolution and parameterizations. For example, the dominance of sedimentary Nd in the Pacific may be amplified in regions where OCIMv2's coarse grid under-represents boundary exchange, and future work coupling optimized Nd sources with transport matrices from higher-resolution ocean models could help bridge these gaps. Finally, our model applies a uniform global concentration for aeolian Nd inputs. While this provides a useful first-order approximation, the use of a spatially

variable dust source term in future work could refine our understanding of the relative strength of top-down versus bottom-up controls on the Nd distribution, particularly in regions far from boundary sources.

While GNOM does not include a parameterization of sedimentary diagenesis and thus cannot explicitly test for diagenetic overprinting of ϵNd records, it does inform us on where this is most likely to occur. Our optimized Best Fit model favored increasing reactivity of sediments at both extremely high and low sedimentary ϵNd (Figure S3d in Supporting Information S1), revealing a mechanism that is important for reproducing observed seawater ϵNd values. We thus propose that diagenetic overprinting is most likely to occur where sediments have a combination of high Nd concentrations and extreme ϵNd , such as the North Pacific rim and the Labrador Sea (Blaser et al., 2020; Du et al., 2016).

4. Conclusions

Our model results underscore the challenges of using ϵNd as a tracer of past ocean circulation, while also paving the way toward its broader application in different ocean regimes. The dominant source of Nd to the Atlantic Ocean is at the surface (mostly rivers), thus ϵNd records in the Atlantic Ocean can be interpreted in the context of the top-down hypothesis and be used as a proxy for water-mass mixing. Conversely, the Pacific Ocean is dominated by Nd from sedimentary sources from all depths. The Pacific basin is thus primarily a bottom-up regime where ϵNd paleorecords require more careful interpretations. In areas of the Pacific where there is low sediment reactivity, such as the center of the basin, variability in ϵNd records may reflect changes in deep water residence time. Near the continental margins of the Pacific, ϵNd records may be subjected to diagenetic overprinting due to the more reactive nature of shallower sediments. We conclude by acknowledging that our use of a coarse-resolution, data-constrained model, while computationally efficient and consistent with other global Nd cycle models, inherently simplifies key processes such as boundary exchange and fine-scale mixing. Consequently, our results represent a robust yet broad-scale view, and we anticipate that the hypotheses generated here will be refined by future high-resolution modeling that can more explicitly capture sediment-water interactions and coastal dynamics.

In addition, this work improves our ability to select locations for paleoceanographic reconstructions. Expanding these records to better understand how ocean circulation changed during major climate shifts in the past is paramount for anticipating and preparing for the impacts of anthropogenic climate change.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

The latest version of GNOM (Pasquier, Hines, et al., 2022; Pasquier, Primeau, et al., 2022) used to perform diagnostic and sedimentary flux experiments is available via Pasquier et al. (2023) and is developed openly on Github.

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